Setup system

Setup the system and get files from the internet.

In the following code, we will download the necessary files from the internet.

Make videos I: Pick Find character images

Assignment: Find character images

1. Click on your Folder icon on the Left Hand Side.
2. Click on `drive`.
3. Click on `MyDrive`.
4. Click on `Race-2025`.
5. Click on `GraphSpeeds2-main`.
6. Look at the different images.
7. Write down the names of the image files that you want to use.

Make videos II: Create tables for your characters

Assignment: Create the tables and equations for your characters

1. Enter time and distance units in `column_labels`.
2. Enter `time_values` and `distance_values` for each value.
3. Enter names and equations for each character.

In [2]:
```
# Prepare the labels for everyone
column_labels = ["x (seconds)", "y (feet)"] # <- FIX units

os.chdir("/content/drive/MyDrive/Race-2025/GraphSpeeds2-main")
!wget https://raw.githubusercontent.com/pattichis/GraphSpeeds/main/Koala.jpeg

# Table #1 values:
time_values     = [0,1,2,6.009,12]  # <- ENTER x values
distance_values = [0, 4.16, 8.32, 25,50]  # <- ENTER y values

# Create your table:
char1 = table(x=time_values, y=distance_values, column_labels=column_labels)
char1.plotTable()
char1.saveImage("char1.png")


# Table #2 values erin:
time_values     = [0,1,2,3.5098,7 ]  # <- ENTER x values
distance_values = [0, 7.14, 14.28, 25,50]  # <- ENTER y values

# Create your table:
char2 = table(x=time_values, y=distance_values, column_labels=column_labels)
char2.plotTable()
```

```
char2.saveImage("char2.png")



# Table #3 values:
time_values     = [0, 1,2,4, 8]  # <- ENTER x values
distance_values = [0, 6.25, 12.5, 25,50]  # <- ENTER y values

# Create your table:
char3 = table(x=time_values, y=distance_values, column_labels=column_labels)
char3.plotTable()
char3.saveImage("char3.png")



# Table #4 values:
time_values     = [0, 1, 2, 2.5,5]  # <- ENTER x values
distance_values = [0, 10, 20, 25,50]  # <- ENTER y values

# Create your table:
char4 = table(x=time_values, y=distance_values, column_labels=column_labels)
char4.plotTable()
char4.saveImage("char4.png")


# Merge all of the tables
tables_list = [char1, char2, char3, char4]

# Line equations of tables
legend_labels = [
      'hello kitty: y = 4.16*x', # <- Modify name and the equation here.
      'shadow milk: y = 7.14*x', # <- Modify name and the equation here.
      'koala: y = 6.25*x', # <- Modify name and the equation here.
      'george: y = 10*x'] # <- Modify name and the equation here.

# Plot tables
plotTablesLines(tables_list,
                fig_title = 'The Race',               # title on top
                x_label = 'Elapsed time (seconds)',    # appears along x-axis
                y_label = 'Distance in feet',       # appears along y-axis
                legend_title = 'Journey',            # title on the right hand side
                legend_labels = legend_labels,       # text describing the table
                img_name = 'race.png')               # Saves under race.png.
```

```
--2026-08-04 22:55:13--  https://raw.githubusercontent.com/pattichis/GraphSpeeds/main/Koala.jpeg
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.109.133
, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443... connect
ed.
HTTP request sent, awaiting response... 200 OK
Length: 205381 (201K) [image/jpeg]
Saving to: ‘Koala.jpeg.3’


Koala.jpeg.3          0%[                    ]       0  --.-KB/s
Koala.jpeg.3        100%[===================>] 200.57K  --.-KB/s    in 0.03s

2026-08-04 22:55:13 (7.03 MB/s) - ‘Koala.jpeg.3’ saved [205381/205381]
```

x (seconds)	y (feet)
0.00000	0.00000
1.00000	4.16000
2.00000	8.32000
6.00900	25.00000
12.00000	50.00000

```
Wrote  char1.png
```

x (seconds)	y (feet)
0.00000	0.00000
1.00000	7.14000
2.00000	14.28000
3.50980	25.00000
7.00000	50.00000

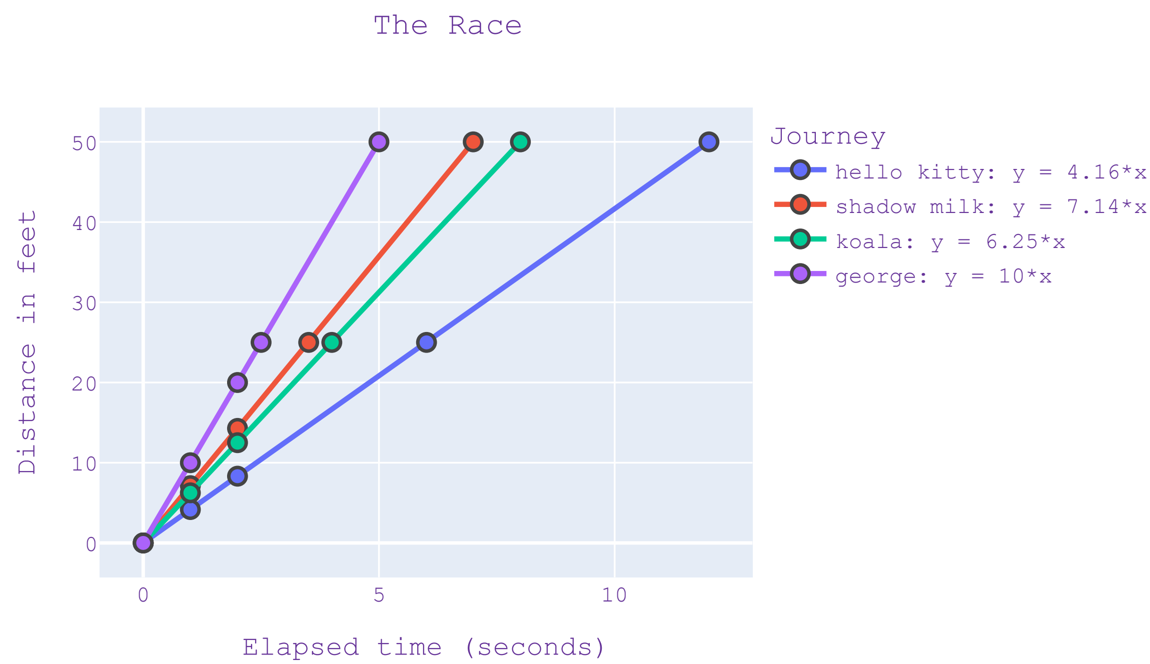
```
Wrote  char2.png
```

x (seconds)	y (feet)
0.00000	0.00000
1.00000	6.25000
2.00000	12.50000
4.00000	25.00000
8.00000	50.00000

```
Wrote  char3.png
```

x (seconds)	y (feet)
0.00000	0.00000
1.00000	10.00000
2.00000	20.00000
2.50000	25.00000
5.00000	50.00000

```
Wrote  char4.png
```



Make videos III: Specify names, images, and speeds

Assignment: place your characters

1. ENTER the names of the characters for your video.
2. ENTER filenames for each character in your video.
3. ENTER the speeds for your characters.

In [3]:
```
# Enter the names of your characters:
names  = ["hello kitty", "shadow milk", "koala", "curious george"]

# Enter the file names of your images:
images = ["/content/drive/MyDrive/Race-2025/GraphSpeeds2-main/hello-kitty-2.png",
          "/content/drive/MyDrive/Race-2025/GraphSpeeds2-main/Shadow_milk_sing.png",
          "/content/drive/MyDrive/Race-2025/Koala.jpeg" ,
          "/content/drive/MyDrive/Race-2025/GraphSpeeds2-main/Curious-George.jpeg"]

# Enter the speeds:
speeds = [4.16, 7.14, 6.25,  10]


# Directory:
directory = ""
images_list = [directory + filename for filename in images]

# Save all of the data:
data = [names, images_list, speeds]
```

Make videos IV: Make and experiment with your video

Your video will be stored to the video file `race.mp4` or a filename that you choose.

To make a video, one time unit (e.g., one hour) corresponds to one second of video.

If you set the `simulation_speed=2.0`, then one second will correspond to two time units (e.g., two hours).

For each second, the video contains a number of images. This is controlled by `fps` which means number of frames per second.

Assignment: experiment with your video

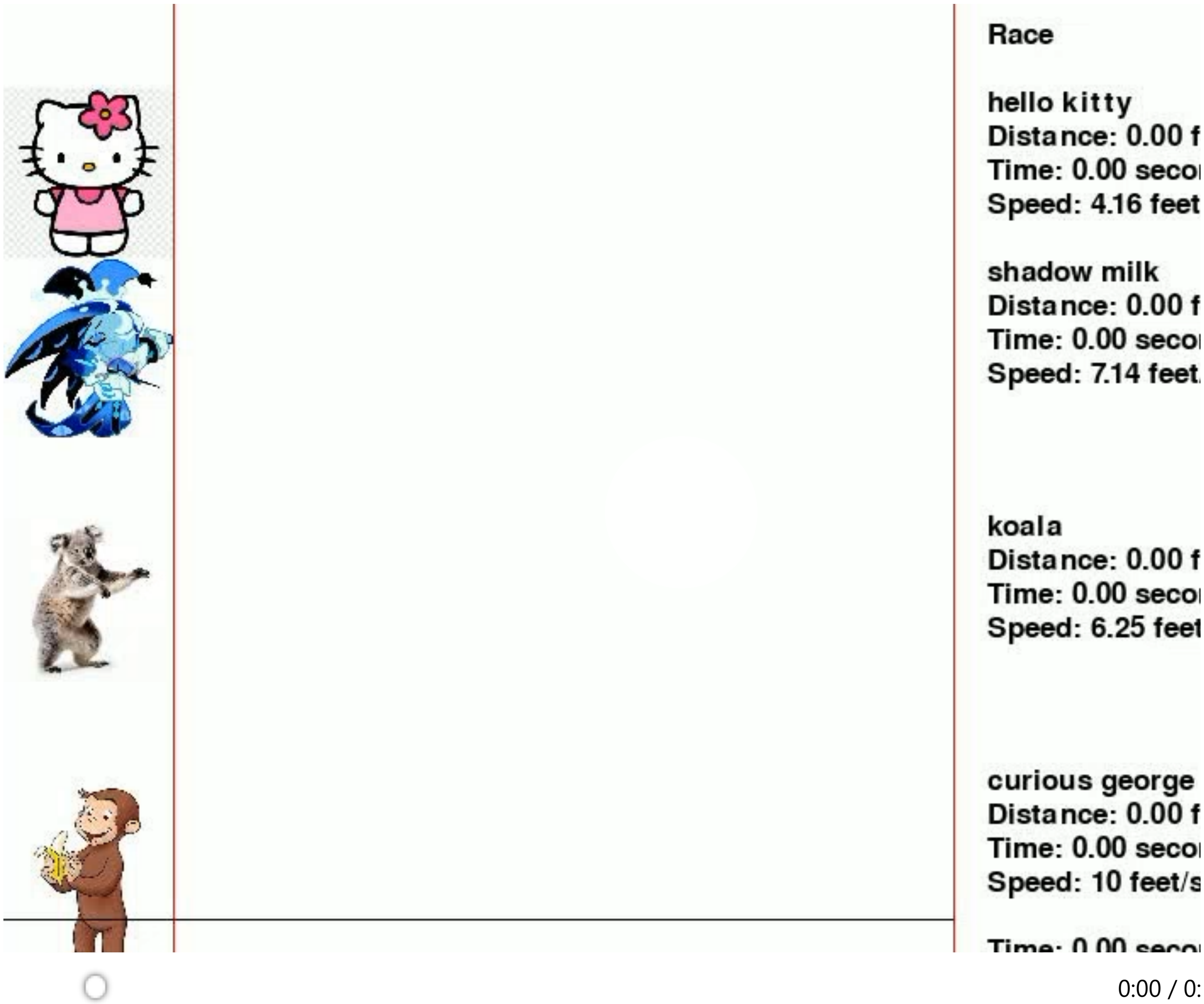
1. Enter the race distance, duration of the race, and video title.
2. Enter the units.
3. Change `simulation_speed` to 2.0. What do you observe?
4. Change `fps` to 20. What do you observe?
5. Save the video to a different file name.

In [4]:

```
# Experiment with the video
sim_video = simulationVideo(data = data,            # characters
                             duration = 15,         # ENTER duration
                             race_distance = 50,    # ENTER race distance
                             vid_title = "Race")    # video title

sim_video.set_video(video_name="race.mp4",          # new filename
                    fps=10,                         # frames per second
                    simulation_speed=1.0)           # speed of the simulation

sim_video.set_units(distance_string="feet",         # ENTER distance unit.
                    time_string="seconds",          # ENTER time unit.
                    speed_string="feet/second")     # ENTER speed units.


# Create and save your video:
video = sim_video.create_video()
video.ipython_display()
```

```
File 'race.mp4' has been removed.
VideoWriter initialized successfully.
video file =  race.mp4  closed.
Moviepy - Building video __temp__.mp4.
Moviepy - Writing video __temp__.mp4


Moviepy - Done !
Moviepy - video ready __temp__.mp4
```

Out[4]:



Make videos V: Make title and explanatory text images

You can create many text frames and save them into different frames.
This is done using `saveImage("filename.png")` as given below:

```
text = """First line.
Second line.
Third line."""
textImage(frame1.png", text)
```

Assignment: Create text images

Fill out the explanatory text images.

In [5]:
```
# Create Title and group name here
text = """Session 1 Group B:
Team guitar.
"""
textImage("frame1.png", text)

# Who is racing?
```

```
text = """ The hello kitty and goerge,the koala,and shadow milk are racing.
"""
textImage("frame2.png", text)

# Why they are racing?
text = """ They are racing because they're fighting over a jacket.
...
"""
textImage("frame3.png", text)


# Character 1 equation and explanation:
text = """ hello kitty equation:
y = 4.16 * x
it was going 4.16 ft per second
"""
textImage("frame4.png", text)

# Character 2 equation and explanation:
text = """ shadow milk equation:
y = 7.14 * x
it means that shadow milk was going the constant speed of 7.14
"""
textImage("frame5.png", text)

# Character 3 equation and explanation:
text = """ koala equation:
y = 6.25 * x
it means that the koala was going 6.25 ft per second.
"""
textImage("frame6.png", text)


# Character 4 equation and explanation:
text = """ george equation:
y = 10 * x
it means that George was going 10 ft per second.
"""
textImage("frame7.png", text)


# Explain the winner:
text = """ The winner is curious george becasue he went 10 feet per second.
...
"""
textImage("frame8.png", text)

# Victory page:
text = """ the end
...
"""
textImage("frame9.png", text)
```

Make videos VII: Make the final video

You can make the final video by combining images and videos together.

Assignment: Make final video

Verify that your video satisfies the requirements of your assignment!

In [6]:
```python
# Confirm that you have all the image here:
img_list = ["frame1.png", "frame2.png", "frame3.png",
            "race.mp4",
            "char1.png", "frame4.png",
            "char2.png", "frame5.png",
            "char3.png", "frame6.png",
            "char4.png", "frame7.png", "race.png",
            "frame8.png", "frame9.png"]

# Create the video:
video=CreateVideo(video_name="all.mp4",     # This is file name of the final video.
                  file_list=img_list,        # This the list of images and files to combine.
                  fps=10,                   # The number of frames per second.
                  duration=2.0)            # The duration for each image.

# Display the video:
video.ipython_display()
```

```
Opening  frame1.png
Opening  frame2.png
Opening  frame3.png
Opening  race.mp4
Opening  char1.png
Opening  frame4.png
Opening  char2.png
Opening  frame5.png
Opening  char3.png
Opening  frame6.png
Opening  char4.png
Opening  frame7.png
Opening  race.png
Opening  frame8.png
Opening  frame9.png
video: height =  600  width =  800
File 'all.mp4' has been removed.
VideoWriter initialized successfully.
Moviepy - Building video __temp__.mp4.
Moviepy - Writing video __temp__.mp4



Moviepy - Done !
Moviepy - video ready __temp__.mp4
```

Out[6]:

0:00 / 0:

SAVE YOUR VIDEOS AND GOOGLE COLAB NOTEBOOKS

In [11]:

```
!jupyter nbconvert --to html /content/drive/MyDrive/Colab\ Notebooks/GraphSpeeds_C1_G2.ipynb
```

```
[NbConvertApp] Converting notebook /content/drive/MyDrive/Colab Notebooks/GraphSpeeds_C1_G2.ipynb t
o html
[NbConvertApp] Writing 3824128 bytes to /content/drive/MyDrive/Colab Notebooks/GraphSpeeds_C1_G2.ht
ml
```